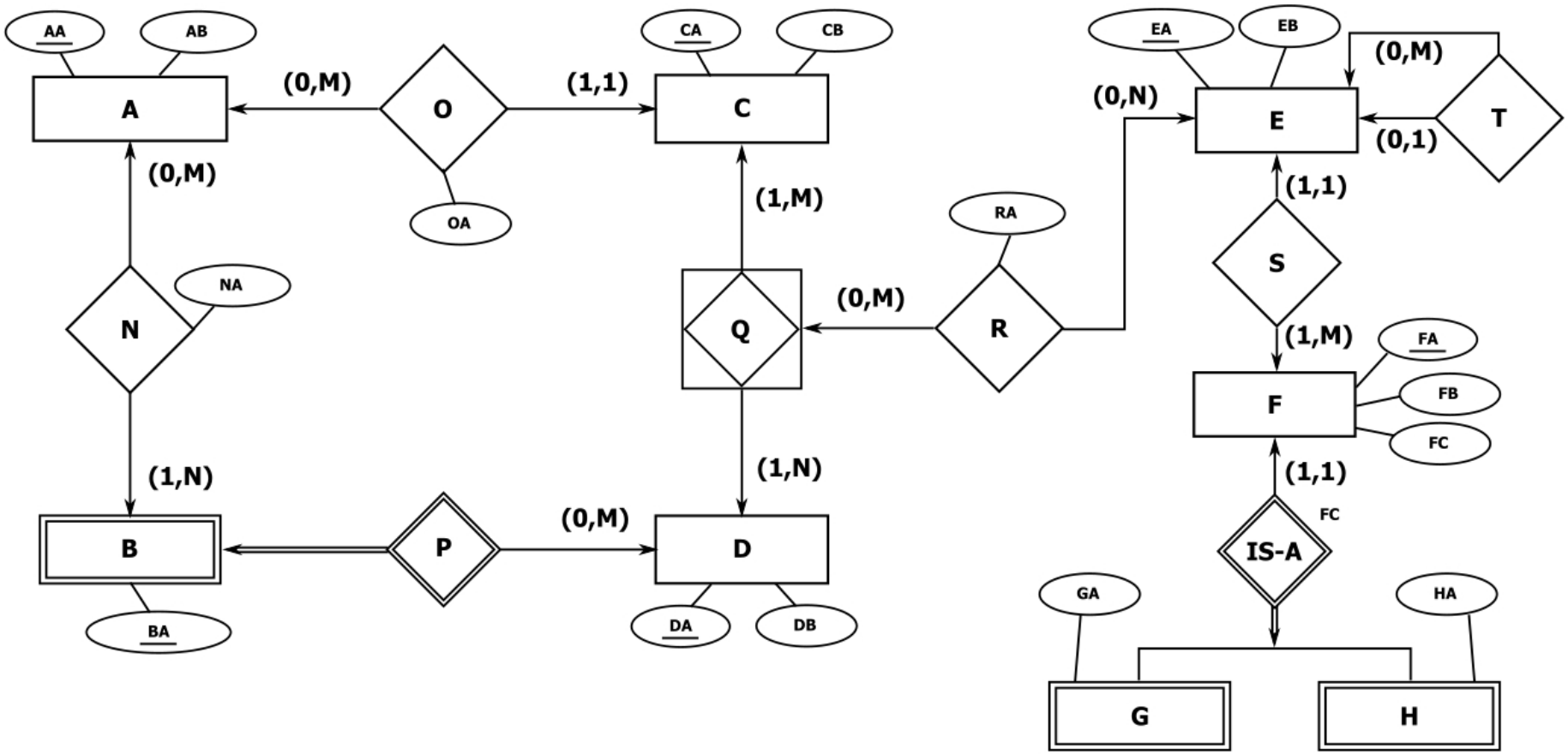
# Prevod ER šeme u relacionu šemu baze podataka

**Baze podataka 2 (2021/2022)**
**OAS Primenjeno softversko inženjerstvo**
**Fakultet tehničkih nauka**
**Univerzitet u Novom Sadu**

# Zadaci

- ZADATAK 1

  Prevesti datu ER šemu u relacionu šemu baze podataka!



# Zadaci

- ZADATAK 1

  A({AA, AB}, {AA})

  C({CA, CB, AA, OA}, {CA})
  C[AA] $\subseteq$ A[AA]
  Null(C, AA) = $\perp$

  D({DA, DB}, {DA})

  B({DA, BA}, {DA+BA})
  B[DA] $\subseteq$ D[DA]

  N({AA, DA, BA, NA}, {AA+DA+BA})
  N[AA] $\subseteq$ A[AA]
  N[DA+BA] $\subseteq$ B[DA+BA]
  B[DA+BA] $\subseteq$ N[DA+BA]

# Zadaci

- ZADATAK 1

  $Q(\{CA, DA\}, \{CA+DA\})$
  $Q[CA] \subseteq C[CA]$
  $C[CA] \subseteq Q[CA]$
  $Q[DA] \subseteq D[DA]$
  $D[DA] \subseteq Q[DA]$

  $F(\{FA, FB, FC\}, \{FA\})$
  $G(\{FA, GA\}, \{FA\})$
  $G[FA] \subseteq F[FA]$
  $H(\{FA, HA\}, \{FA\})$
  $H[FA] \subseteq F[FA]$
  $F[FA] \subseteq G[FA] \cup H[FA]$
  $G[FA] \cap H[FA] = \emptyset$

# Zadaci

- ZADATAK 1

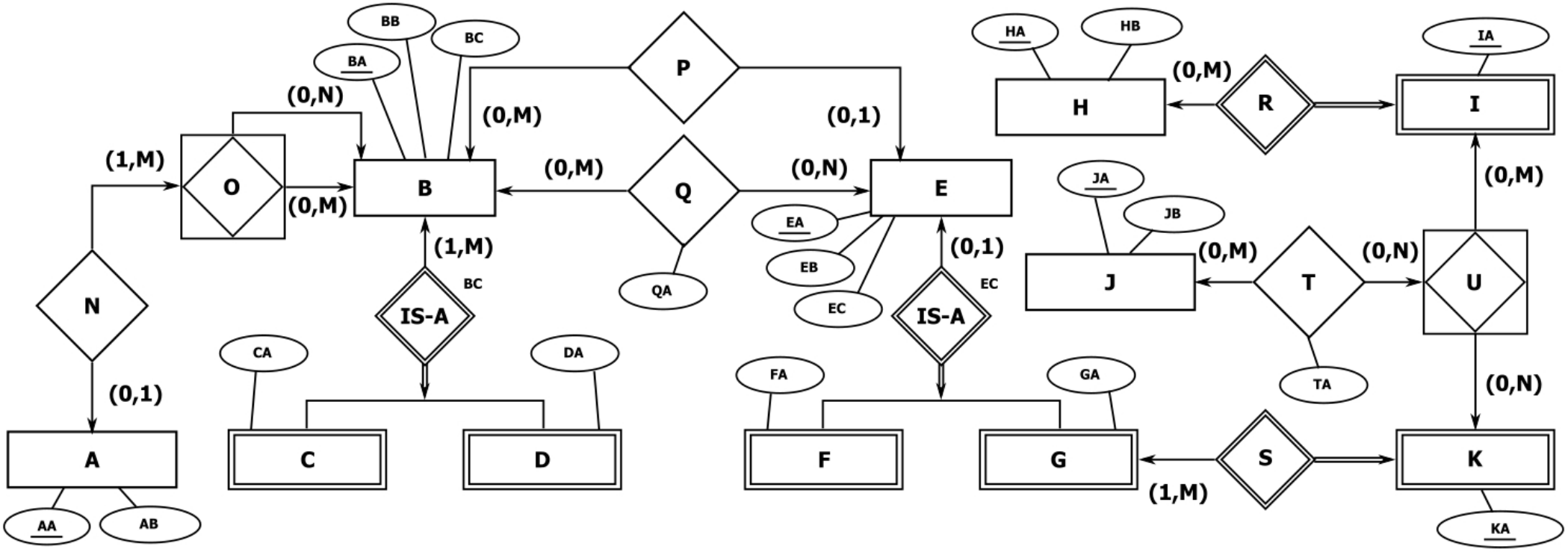
  E({EA, EB, FA, EAX}, {EA})
  E[FA] $\subseteq$ F[FA]
  Null(E, FA) = $\perp$
  F[FA] $\subseteq$ E[FA]
  dom(EAX) $\subseteq$ dom(EA)
  E[EAX] $\subseteq$ E[EA]
  Null(E, EAX) = $\top$

  R({CA, DA, EA, RA}, {CA+DA+EA})
  R[CA+DA] $\subseteq$ Q[CA+DA]
  R[EA] $\subseteq$ E[EA]

# Zadaci

- ZADATAK 2

  Prevesti datu ER šemu u relacionu šemu baze podataka!



# Zadaci

- ZADATAK 2

  B({BA, BB, BC}, {BA})
  C({BA, CA}, {BA})
  C[BA] ⊆ B[BA]
  D({BA, DA}, {BA})
  D[BA] ⊆ B[BA]
  B[BA] ⊆ C[BA] ∪ D[BA]

  O({BA, BAX}, {BA+BAX})
  O[BA] ⊆ B[BA]
  dom(BAX) ⊆ dom(BA)
  O[BAX] ⊆ B[BA]

# Zadaci

- ZADATAK 2

  A({AA, AB, BA, BAX}, {AA})
  A[BA+BAX] $\subseteq$ O[BA+BAX]
  Null(A, BA) = $\top$
  Null(A, BAX) = $\top$
  O[BA+BAX] $\subseteq$ A[BA+BAX]

  E({EA, EB, EC, BA}, {EA})
  F({EA, FA}, {EA})
  F[EA] $\subseteq$ E[EA]
  G({EA, GA}, {EA})
  G[EA] $\subseteq$ E[EA]
  F[EA] $\cap$ G[EA] = $\emptyset$
  E[BA] $\subseteq$ B[BA]
  Null(E, BA) = $\top$

# Zadaci

- ZADATAK 2

  Q({BA, EAX, QA}, {BA+EAX})
  Q[BA] ⊆ B[BA]
  dom(EAX) ⊆ dom(EA)
  Q[EAX] ⊆ E[EA]

  H({HA, HB}, {HA})

  I({HA, IA}, {HA+IA})
  I[HA] ⊆ H[HA]

  K({EA, KA}, {EA+KA})
  K[EA] ⊆ G[EA]
  G[EA] ⊆ K[EA]

# Zadaci

- ZADATAK 2

  U({HA, IA, EA, KA}, {HA+IA+EA+KA})
  U[HA+IA] ⊆ I[HA+IA]
  U[EA+KA] ⊆ K[EA+KA]

  J({JA, JB}, {JA})

  T({JA, HA, IA, EA, KA, TA}, {JA+HA+IA+EA+KA})
  T[JA] ⊆ J[JA]
  T[HA+IA+EA+KA] ⊆ U[HA+IA+EA+KA]

# Literatura


- Pavle Mogin, Ivan Luković, Miro Govedarica. Principi projektovanja baza podataka. 2. izdanje. Fakultet tehničkih nauka, Novi Sad. 2004.